

Answer to the Question No 1

The system takes 4 bit binary system.

$$1 = 0001$$

$$2 = 0010$$

$$3 = 0011$$

$$4 = 0100$$

$$5 = 0101$$

$$6 = 0110$$

$$7 = 0111$$

$$8 = 1000$$

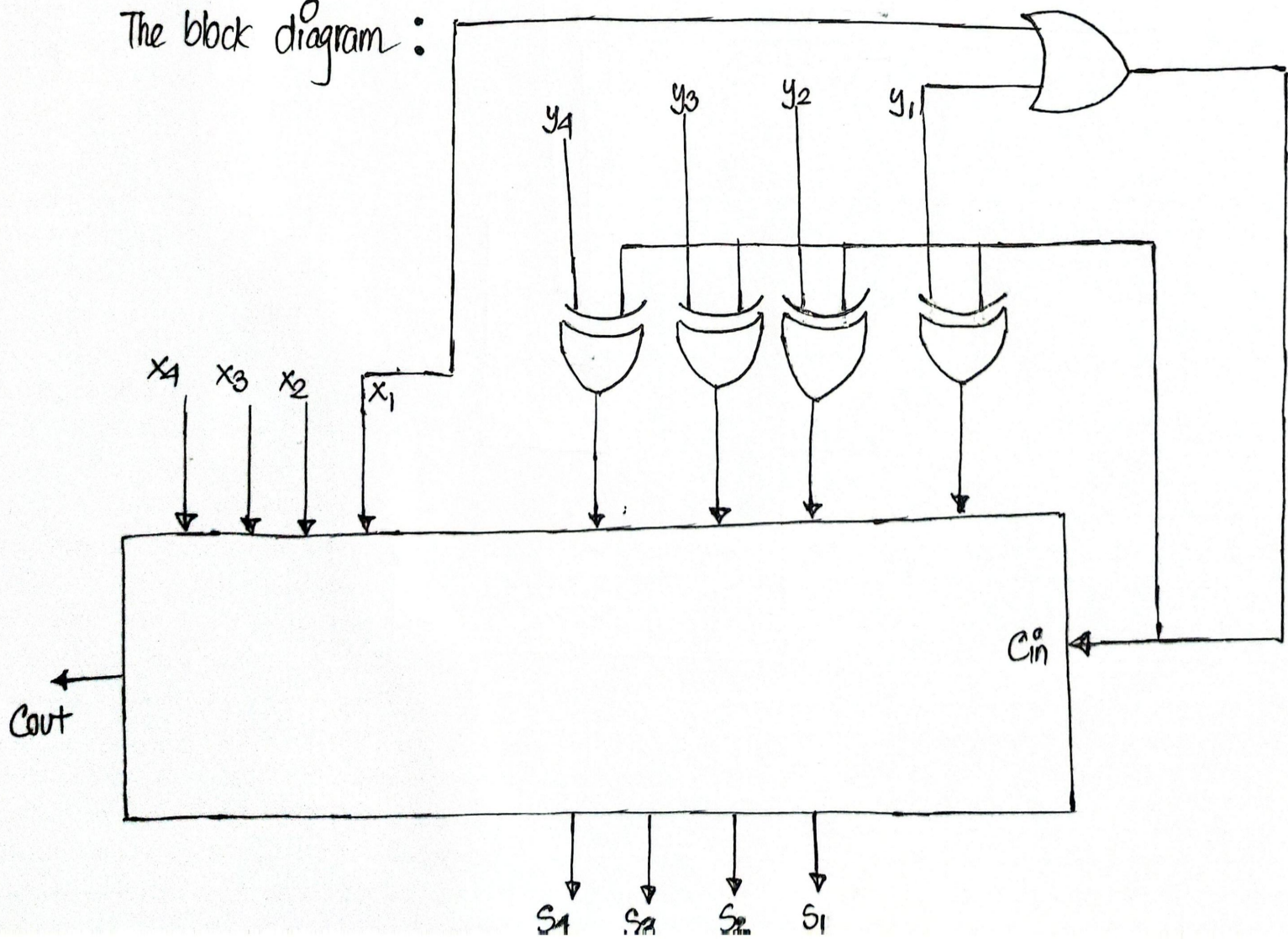
$$A = x_4 x_3 x_2 x_1$$

$$y = y_4 y_3 y_2 y_1$$
$$0 \ 1 \ 0 \ 0$$

For odd numbers, $LSB = 1$

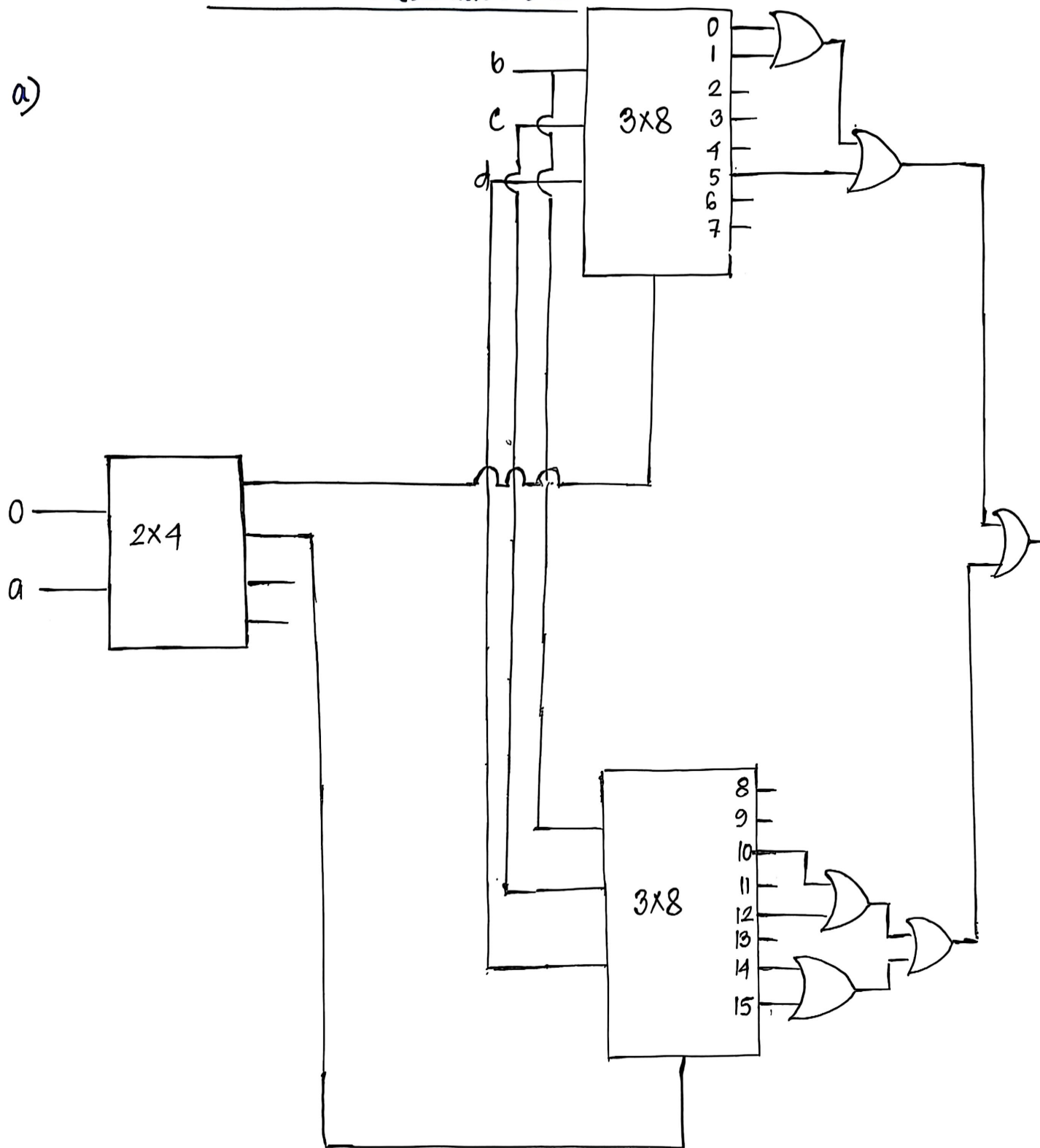
For even numbers, $LSB = 0$

The block diagram :



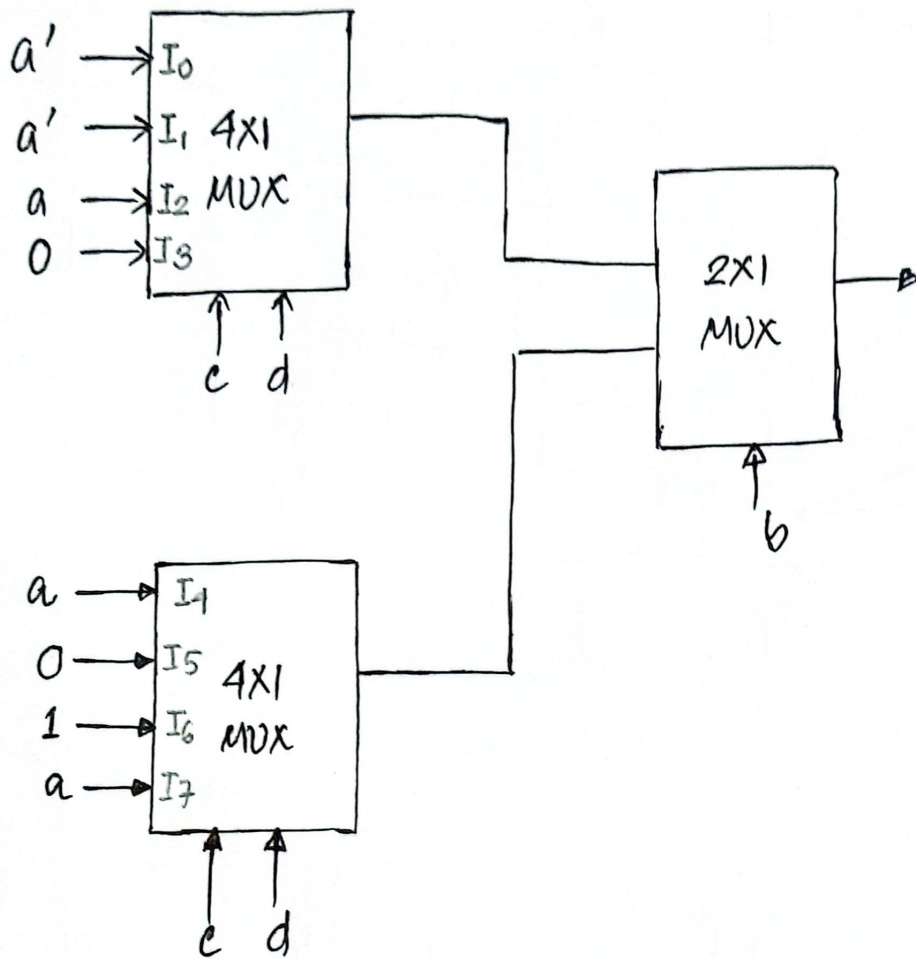
Answer to the Question No 2

a)



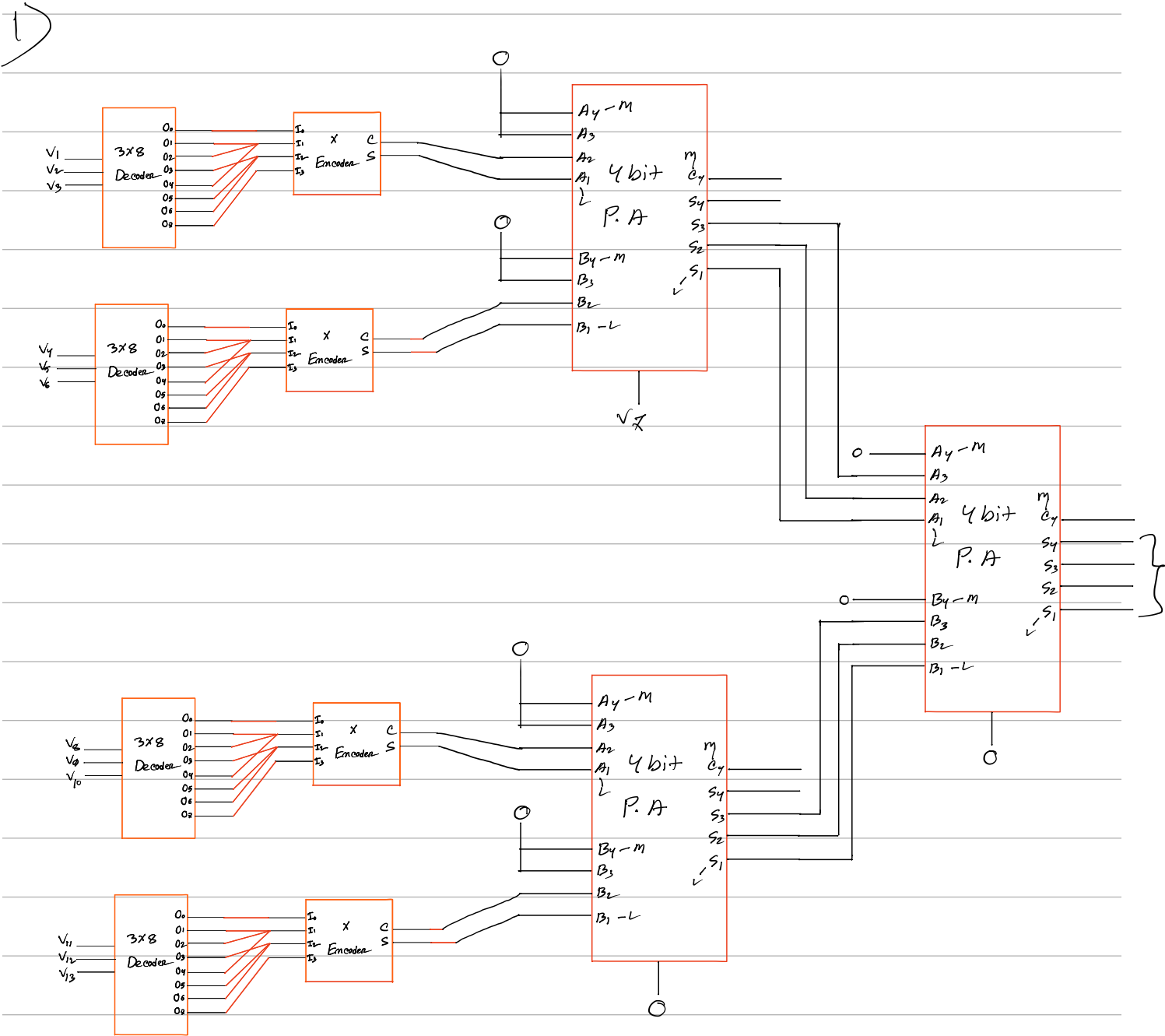
b)

$$F(a, b, c, d) = \sum (0, 1, 6, 10, 12, 14, 15)$$



	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
a'	0	1	2	3	4	5	6	7
a	8	9	10	11	12	13	14	15
	a'	a'	a	0	a	0	1	a

A3 Solve



✓

	D_3	D_2	D_1	D_0	X_1	X_0	
0	0	0	0	0	x	x	✓
1	0	0	0	1	0	0	✓
2	0	0	1	0	0	1	✓
3	0	0	1	1	0	1	✓
4	0	1	0	0	1	0	✓
5	0	1	0	1	0	0	✓
6	0	1	1	0	1	0	✓
7	0	1	1	1	1	0	✓
8	1	0	0	0	1	1	✓
9	1	0	0	1	0	0	✓
10	1	0	1	0	0	1	✓
11	1	0	1	1	0	0	
12	1	1	0	0	1	1	✓
13	1	1	0	1	0	0	
14	1	1	1	0	1	1	✓
15	1	1	1	1	1	1	✓

$X_1 = \Sigma(4, 6, 7, 8, 12, 14, 15)$
 $+ d(0)$

$X_2 = \Sigma(2, 3, 8, 10, 12, 14, 15)$
 $+ d(0)$

Use KMap to find the equations and Draw the ckt.

$X_1 = \bar{D}_1 \bar{D}_0 + D_2 D_1$

$X_0 = \bar{D}_3 \bar{D}_2 D_1 + D_3 D_2 D_1 + D_3 \bar{D}_0$